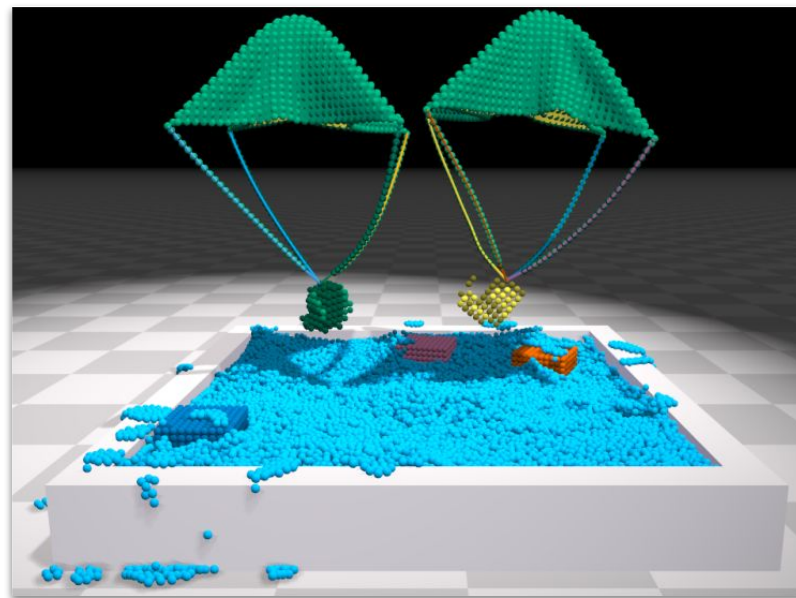
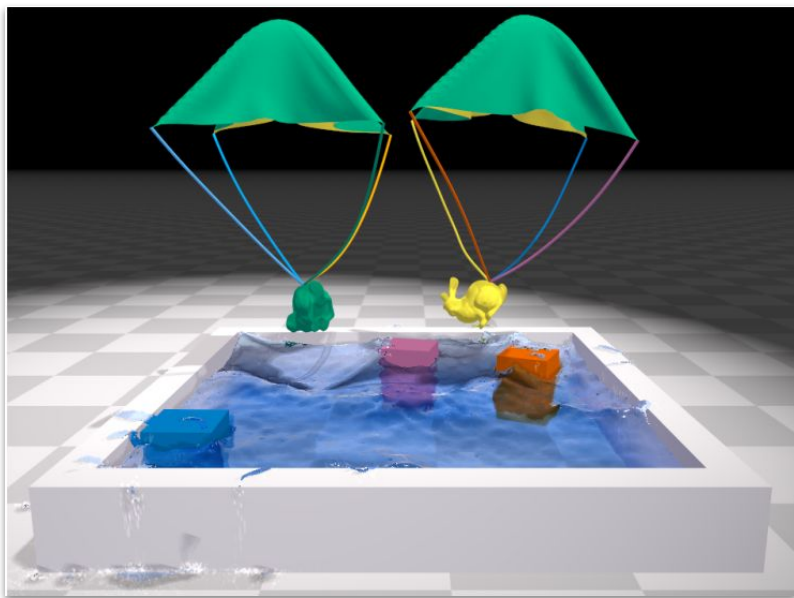


Unified Particle Physics for Real-Time Applications (2014)

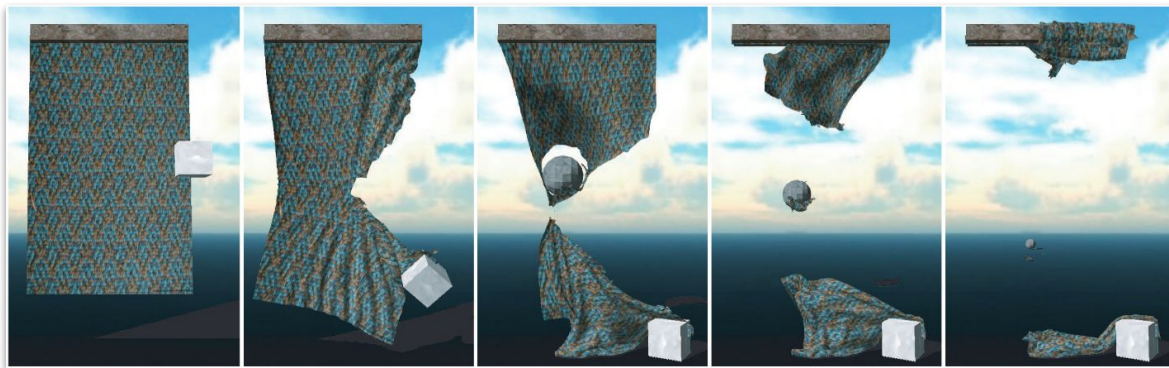
Miles Macklin Matthias Müller Nuttapong Chentanez Tae-Yong Kim



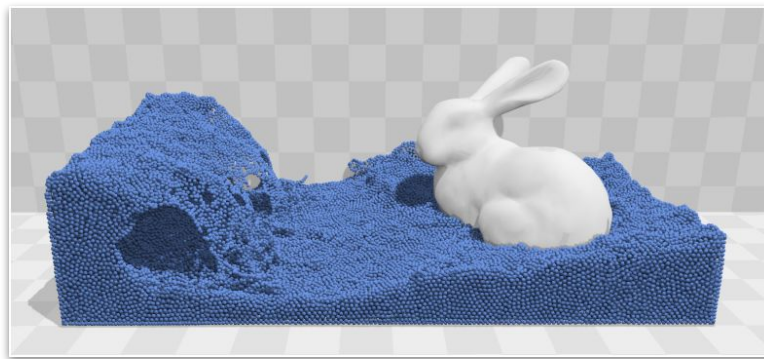
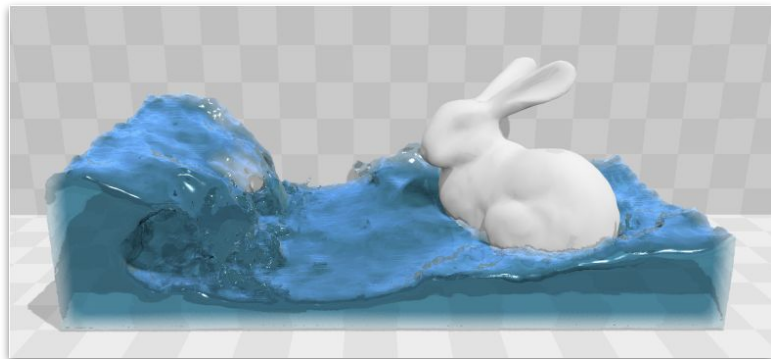
Constraint Based (like PBD)

[illegible]

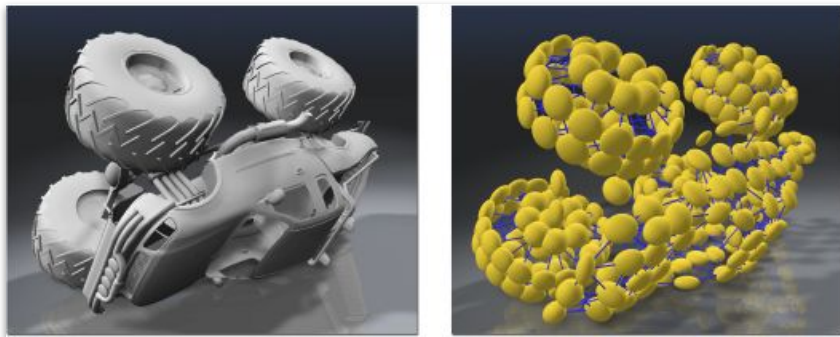
Position Based Dynamic (2006)



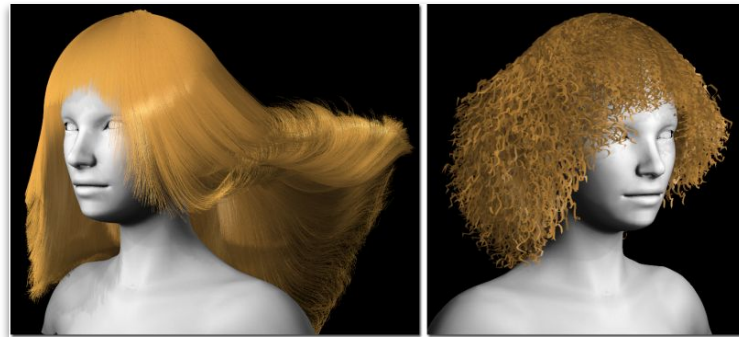
Position Based Fluid (2013)



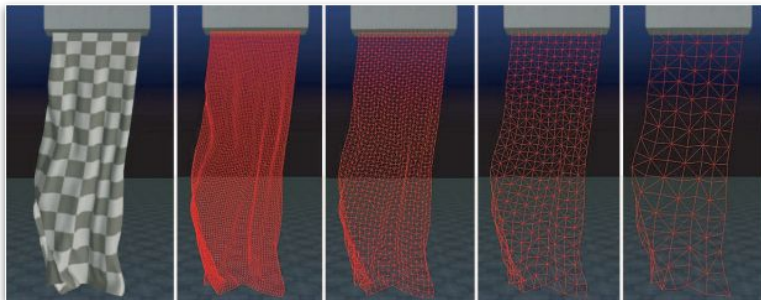
Solid Simulation with Oriented Particles (2011)



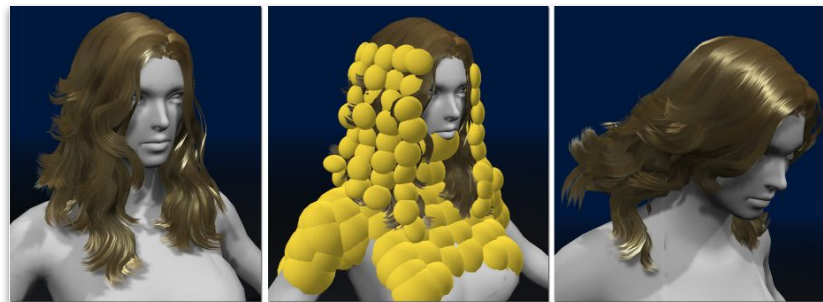
Fast Simulation of Inextensible Hair and Fur (2012)



Hierarchical Position Based Dynamics (2008)



Adding Physics to Animated Characters
with Oriented Particles (2012)



Everything is particles and constraints

```
struct Particle {  
    float x[3]; // position  
    float v[3]; // velocity  
    float inv_mass; // (1 / m)  
    int phase; // stuff  
};
```

Constraints $j \in [1, \dots, M]$

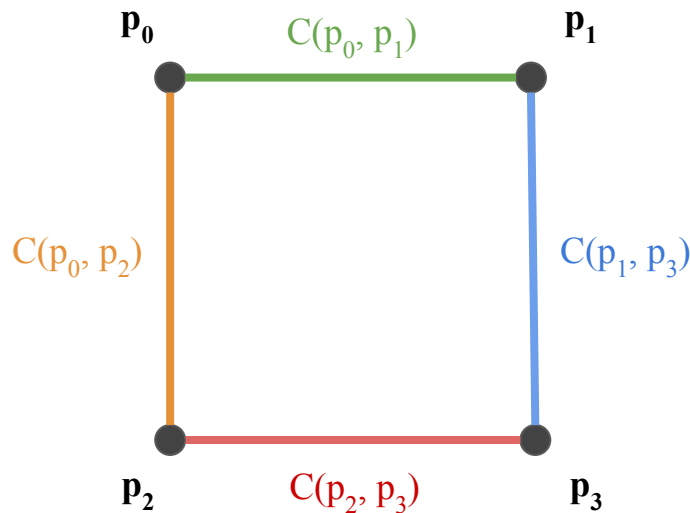
cardinality : n

indices : $\{i_1, \dots, i_{n_j}\}, i_k \in [1, \dots, N]$

fonction : $C_j(\mathbf{x}_{i_1}, \dots, \mathbf{x}_{i_{n_j}}) \rightarrow \mathbb{R}$

stiffness : $k \in [0, 1]$

condition : *equality* ($C = 0$) or *inequality* ($C \geq 0$)



Unified PBD Solver

While(True):

for all particles :

apply forces

predict position

scale mass

$$\mathbf{v}_i \leftarrow \mathbf{v}_i + \Delta t \mathbf{w}_i \mathbf{f}_e$$

$$\mathbf{p}_i \leftarrow \mathbf{x}_i + \Delta t \mathbf{v}_i$$

$$\mathbf{m}_i \leftarrow \mathbf{s}_i \mathbf{m}_i$$

check collisions

for each stabilizationIterations :

solve collisions

collisions stuff

collisions stuff

for each solverIterations :

for all G in constraint_groups :

solve constraints

update predicted position

$$\Delta \mathbf{x}_i \leftarrow \Delta \mathbf{x}_i + \Delta \mathbf{x}_m$$

$$\mathbf{p}_i \leftarrow \mathbf{p}_i + \Delta \mathbf{x}_i (\mathbf{w} / \mathbf{n})$$

for all particles :

update velocity

advect particles

apply internal forces $\mathbf{f}_{\text{drag}}$ $\mathbf{f}_{\text{vort}}$

update position or friction or sleep

$$\mathbf{v}_i \leftarrow (\mathbf{p}_i - \mathbf{x}_i) / \Delta t$$

fluid stuff

fluid forces stuff

$$\mathbf{x}_i \leftarrow \mathbf{p}_i$$

Unified PBD Solver

While(True):

for all particles :

apply forces

predict position

scale mass

$$\mathbf{v}_i \leftarrow \mathbf{v}_i + \Delta t \mathbf{w}_i \mathbf{f}_e$$

$$\mathbf{p}_i \leftarrow \mathbf{x}_i + \Delta t \mathbf{v}_i$$

$$\mathbf{m}_i \leftarrow \mathbf{s}_i \mathbf{m}_i$$

Integration

check collisions

for each stabilizationIterations :

solve collisions

collisions stuff

collisions stuff

for each solverIterations :

for all G in constraint_groups :

solve constraints

update predicted position

$$\Delta \mathbf{x}_i \leftarrow \Delta \mathbf{x}_i + \Delta \mathbf{x}_m$$

$$\mathbf{p}_i \leftarrow \mathbf{p}_i + \Delta \mathbf{x}_i (\mathbf{w} / \mathbf{n})$$

for all particles :

update velocity

advect particles

apply internal forces $\mathbf{f}_{\text{drag}}$ $\mathbf{f}_{\text{vort}}$

update position or friction or sleep

$$\mathbf{v}_i \leftarrow (\mathbf{p}_i - \mathbf{x}_i) / \Delta t$$

fluid stuff

fluid forces stuff

$$\mathbf{x}_i \leftarrow \mathbf{p}_i$$

Unified PBD Solver

While(True):

for all particles :

apply forces

predict position

scale mass

$$\mathbf{v}_i \leftarrow \mathbf{v}_i + \Delta t \mathbf{w}_i \mathbf{f}_e$$

$$\mathbf{p}_i \leftarrow \mathbf{x}_i + \Delta t \mathbf{v}_i$$

$$\mathbf{m}_i \leftarrow \mathbf{s}_i \mathbf{m}_i$$

Integration

check collisions

for each **stabilizationIterations** :

solve collisions

collisions stuff

collisions stuff

Stabilization

for each solverIterations :

for all G in constraint_groups :

solve constraints

update predicted position

$$\Delta \mathbf{x}_i \leftarrow \Delta \mathbf{x}_i + \Delta \mathbf{x}_m$$

$$\mathbf{p}_i \leftarrow \mathbf{p}_i + \Delta \mathbf{x}_i (\mathbf{w} / \mathbf{n})$$

for all particles :

update velocity

advect particles

apply internal forces $\mathbf{f}_{\text{drag}}$ $\mathbf{f}_{\text{vort}}$

update position or friction or sleep

$$\mathbf{v}_i \leftarrow (\mathbf{p}_i - \mathbf{x}_i) / \Delta t$$

fluid stuff

fluid forces stuff

$$\mathbf{x}_i \leftarrow \mathbf{p}_i$$

Unified PBD Solver

While(True):

for all particles :

apply forces

predict position

scale mass

$$\mathbf{v}_i \leftarrow \mathbf{v}_i + \Delta t \mathbf{w}_i \mathbf{f}_e$$

$$\mathbf{p}_i \leftarrow \mathbf{x}_i + \Delta t \mathbf{v}_i$$

$$\mathbf{m}_i \leftarrow \mathbf{s}_i \mathbf{m}_i$$

Integration

check collisions

for each stabilizationIterations :

solve collisions

collisions stuff

collisions stuff

Stabilization

for each solverIterations :

for all $\mathbf{G}$ in constraint_groups :

solve constraints

update predicted position

$$\Delta \mathbf{x}_i \leftarrow \Delta \mathbf{x}_i + \Delta \mathbf{x}_m$$

$$\mathbf{p}_i \leftarrow \mathbf{p}_i + \Delta \mathbf{x}_i (\mathbf{w} / \mathbf{n})$$

Solve Constraints

for all particles :

update velocity

advect particles

apply internal forces $\mathbf{f}_{\text{drag}}$ $\mathbf{f}_{\text{vort}}$

update position or friction or sleep

$$\mathbf{v}_i \leftarrow (\mathbf{p}_i - \mathbf{x}_i) / \Delta t$$

fluid stuff

fluid forces stuff

$$\mathbf{x}_i \leftarrow \mathbf{p}_i$$

Unified PBD Solver

While(True):

for all particles :

apply forces

predict position

scale mass

$$\mathbf{v}_i \leftarrow \mathbf{v}_i + \Delta t \mathbf{w}_i \mathbf{f}_e$$

$$\mathbf{p}_i \leftarrow \mathbf{x}_i + \Delta t \mathbf{v}_i$$

$$\mathbf{m}_i \leftarrow \mathbf{s}_i \mathbf{m}_i$$

Integration

check collisions

for each stabilizationIterations :

solve collisions

collisions stuff

collisions stuff

Stabilization

for each solverIterations :

for all G in constraint_groups :

solve constraints

update predicted position

$$\Delta \mathbf{x}_i \leftarrow \Delta \mathbf{x}_i + \Delta \mathbf{x}_m$$

$$\mathbf{p}_i \leftarrow \mathbf{p}_i + \Delta \mathbf{x}_i (\mathbf{w} / \mathbf{n})$$

Solve Constraints

for all particles :

update velocity

advect particles

apply internal forces $\mathbf{f}_{\text{drag}}$ $\mathbf{f}_{\text{vort}}$

update position or friction or sleep

$$\mathbf{v}_i \leftarrow (\mathbf{p}_i - \mathbf{x}_i) / \Delta t$$

fluid stuff

fluid forces stuff

$$\mathbf{x}_i \leftarrow \mathbf{p}_i$$

Velocity

and

Everything Else

Unified PBD Solver

While(True):

for all particles :

apply forces

predict position

scale mass

$$\mathbf{v}_i \leftarrow \mathbf{v}_i + \Delta t \mathbf{w}_i \mathbf{f}_e$$

$$\mathbf{p}_i \leftarrow \mathbf{x}_i + \Delta t \mathbf{v}_i$$

$$\mathbf{m}_i \leftarrow \mathbf{s}_i \mathbf{m}_i \text{ ————— Stable Stack}$$

check collisions

for each stabilizationIterations :

solve collisions

collisions stuff

collisions stuff

Cuda particles (2010)

SDF + Rigid Collisions

for each solverIterations :

for all G in constraint_groups :

solve constraints

update predicted position

$$\Delta \mathbf{x}_i \leftarrow \Delta \mathbf{x}_i + \Delta \mathbf{x}_m$$

$$\mathbf{p}_i \leftarrow \mathbf{p}_i + \Delta \mathbf{x}_i (\mathbf{w} / \mathbf{n}) \text{ ————— Jacobi Solver}$$

for all particles :

update velocity

advect particles

apply internal forces $\mathbf{f}_{\text{drag}}$ $\mathbf{f}_{\text{vort}}$

update position or friction or sleep

$$\mathbf{v}_i \leftarrow (\mathbf{p}_i - \mathbf{x}_i) / \Delta t$$

fluid stuff

fluid forces stuff

$$\mathbf{x}_i \leftarrow \mathbf{p}_i \text{ ————— Friction ? Sleeping ?}$$

Solving Method

Gauss-Seidel



Solve C1

Diagram illustrating the Gauss-Seidel method. A horizontal line has a red circle on the left and a blue circle on the right. A black dot is in the middle. A red dashed arrow points from the black dot to the red circle. A blue dashed arrow labeled 'C2' points from the black dot to the blue circle.

Solve C2

Diagram illustrating the Gauss-Seidel method. A horizontal line has a red circle on the left and a blue circle on the right. A black dot is in the middle. A red dashed arrow points from the black dot to the red circle. A blue dashed arrow points from the black dot to the blue circle.

...

- + Good convergence rate
- Possible oscillations
- Difficult to parallelize (graph coloring)

Jacobi



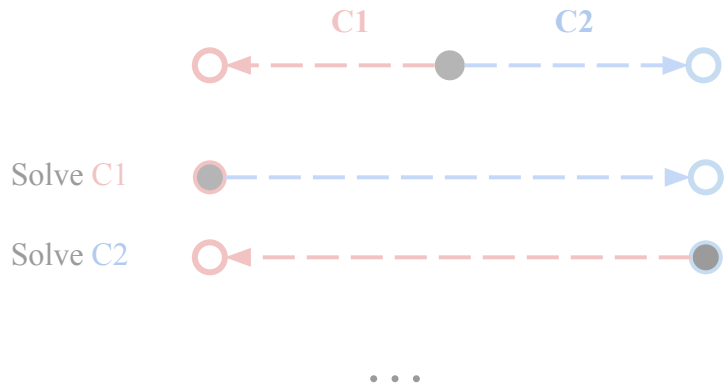
Diagram illustrating the Jacobi method. A horizontal line has a red circle on the left and a blue circle on the right. A black dot is in the middle. A red dashed arrow points from the black dot to the red circle. A blue dashed arrow points from the black dot to the blue circle.

 $(C1 + C2) / 2$

- + GPU friendly
- + Always converge
- Bad convergence rate
- Find good solutions ?

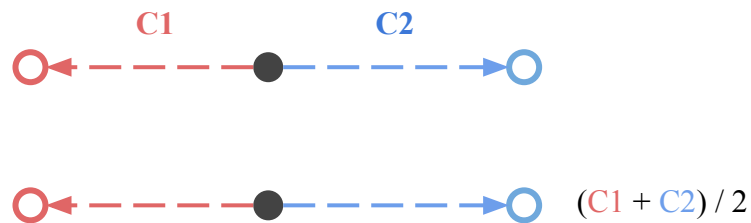
Unified PBD Solver

Gauss-Seidel



- + Good convergence rate
- Possible oscillations
- Difficult to parallelize (graph coloring)

Jacobi



- + GPU friendly
- + Always converge

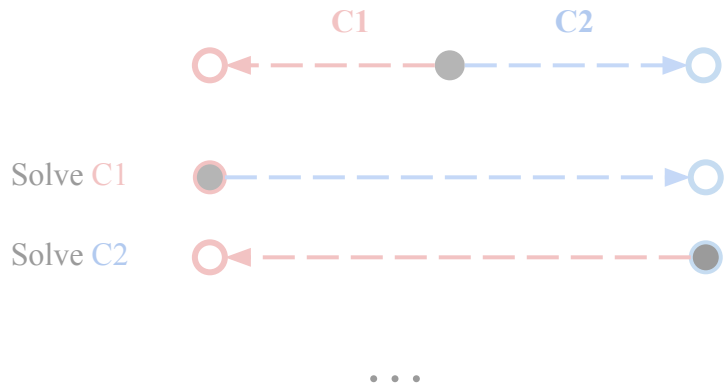
- Bad convergence rate
- Find good solutions ?

$$p_i + \Delta x_i (w / n)$$

$$w \in [1, 2]$$

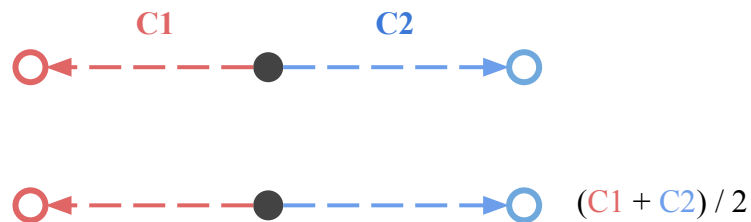
Unified PBD Solver

Gauss-Seidel



- + Good convergence rate
- Possible oscillations
- Difficult to parallelize (graph coloring)

Jacobi



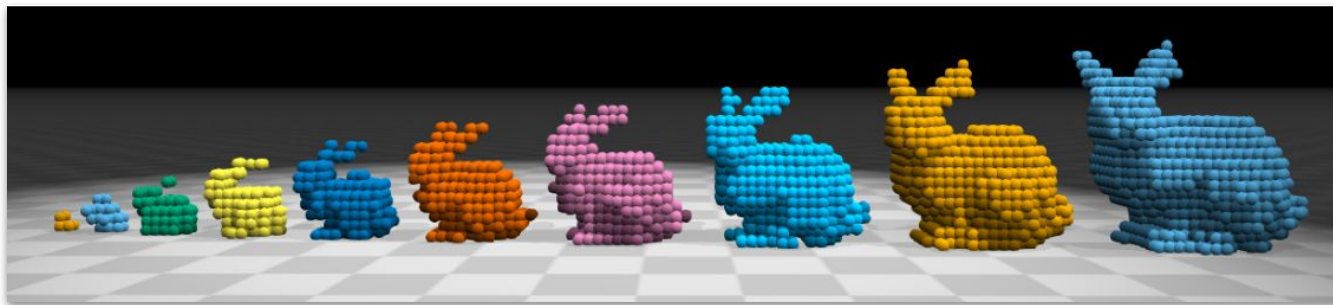
- + GPU friendly
- + Always converge

- Less Bad convergence rate (meh)
- Find good solutions ?

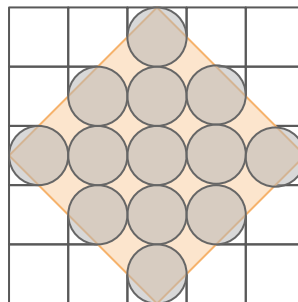
$$p_i + \Delta x_i (w / n)$$

$$w \in [1, 2]$$

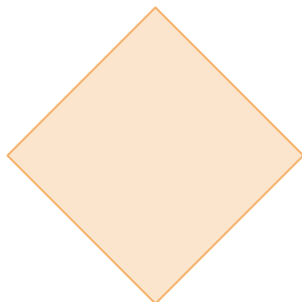
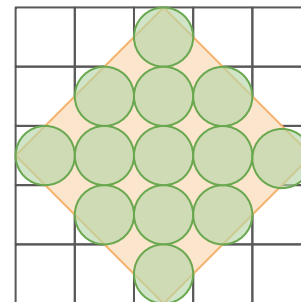
Rigid Body collision



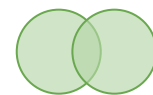
Voxelisation



Assign Phase



Use shape matching constraint !



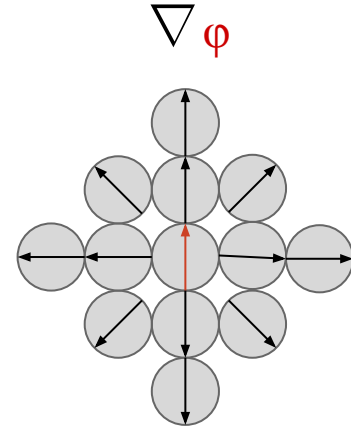
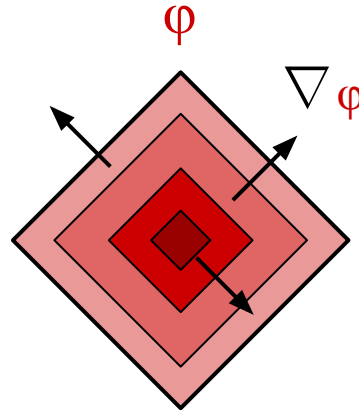
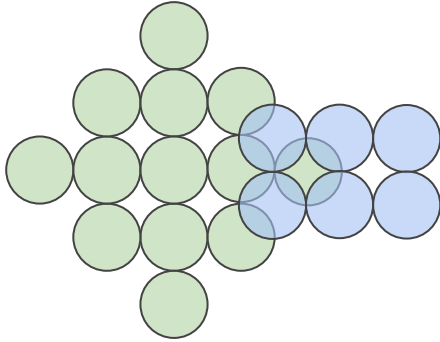
Ignored



Collision !

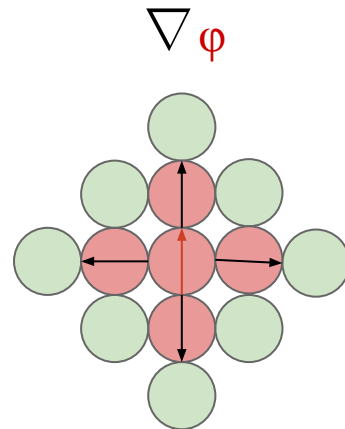
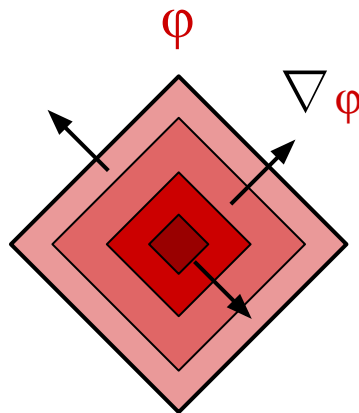
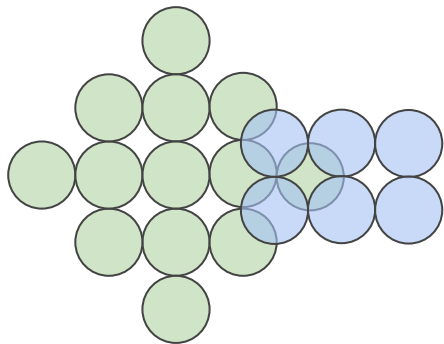
SDF collision

Stuck !



SDF collision

Stuck !



If not **surface** use **SDF** !

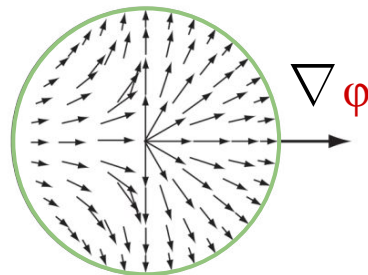
if $r < |\phi|$

$$d = \|x_i - x_j\| - r$$

$$\Delta x_i = \frac{w_1}{w_1 + w_2} (d \cdot n_{ij})$$

$$\Delta x_j = \frac{w_2}{w_1 + w_2} (d \cdot n_{ij})$$

$$n_{ij}^* = \begin{cases} x_{ij} - 2(x_{ij} \cdot n_{ij})n_{ij} & x_{ij} \cdot n_{ij} < 0 \\ x_{ij} & \end{cases}$$

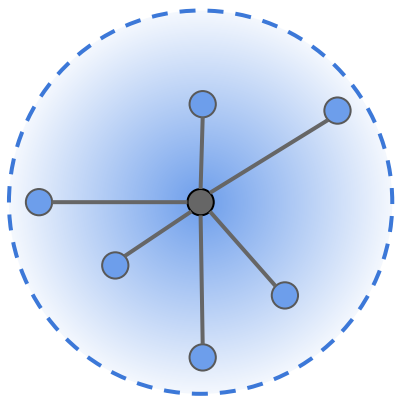


Fluid/Solid Coupling

Fluid constraint (clamped)

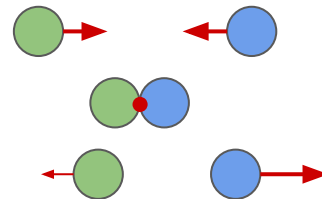
$$C(x) = \frac{\rho_i}{\rho_0} - 1 \leq 0$$

$$\rho_i = \sum_j m_j W(p_i - p_j, h)$$



$$W \in [0,1]$$

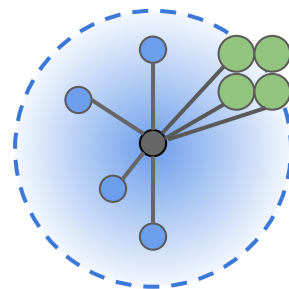
Collisions **Solid** $\Rightarrow$ **Fluid**



Fluid is solid

$$C(x_i, x_j) = \|x_{ij}\| - r \geq 0$$

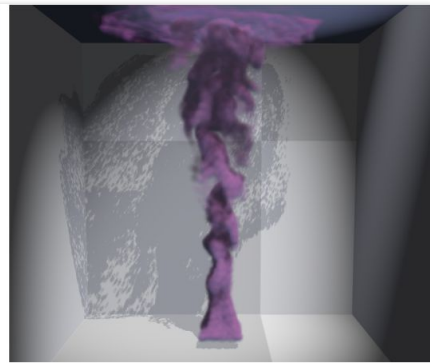
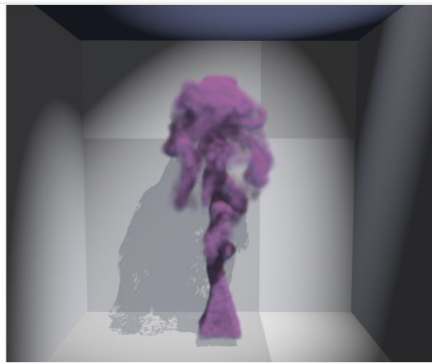
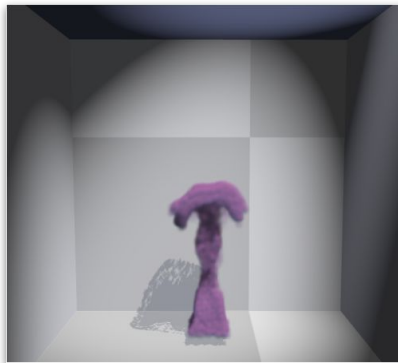
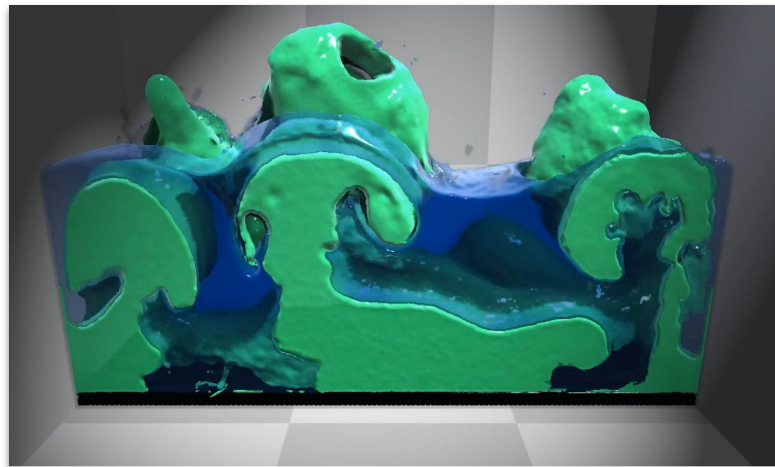
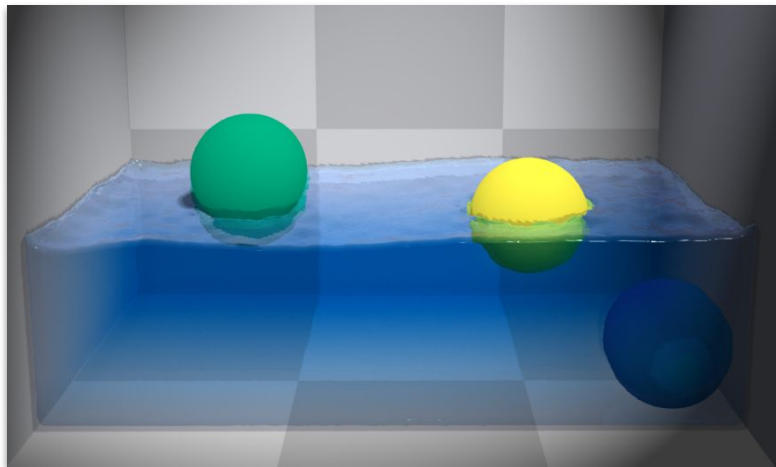
Collisions **Fluid** $\Rightarrow$ **Solid**



Solid is fluid

$$\rho_i = \sum_{\text{fluid}} m_j W(x_i - x_j, h) + s \sum_{\text{solid}} m_k W(x_i - x_j, h)$$

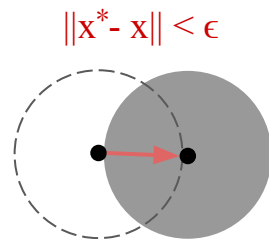
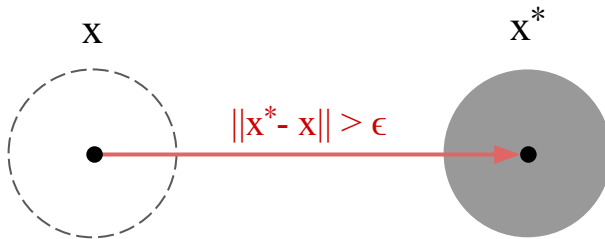
Fluid/Solid Coupling



Friction and sleeping

Particle Sleeping

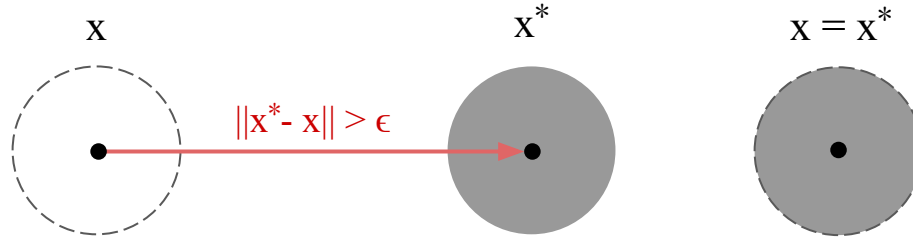
$$\mathbf{x}(t + \Delta t) = \begin{cases} \mathbf{x}^*, & \|\mathbf{x}^* - \mathbf{x}\| > \epsilon \\ \mathbf{x} & \end{cases}$$



Friction and sleeping

Particle Sleeping

$$x(t + \Delta t) = \begin{cases} x^*, & \|x^* - x\| > \epsilon \\ x & \end{cases}$$

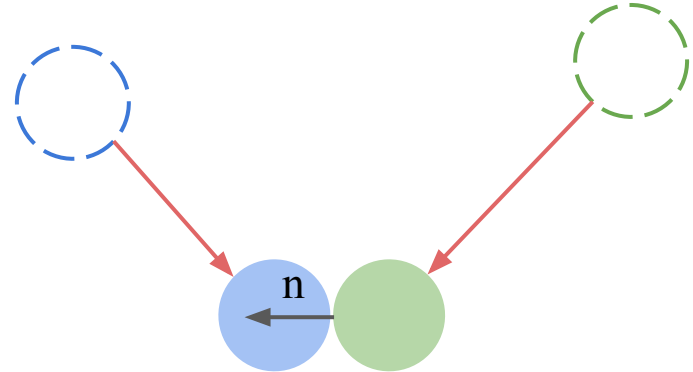


Position Based Friction (Coulomb)

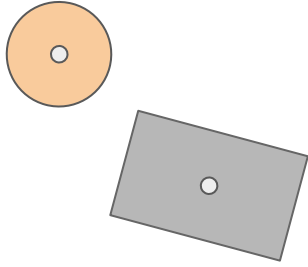
$$\Delta x_{\perp} = [(x_i^* - x_i) - (x_j^* - x_j)] \perp n$$

$$\Delta x_i = \frac{w_1}{w_1 + w_2} \begin{cases} \Delta x_{\perp} & , \|\Delta x_{\perp}\| < \mu_s d \\ \Delta x_{\perp} \min\left(\frac{\mu_k d}{\|\Delta x_{\perp}\|}, 1\right) & \end{cases}$$

μ_k : Kinetic friction μ_s : Static friction

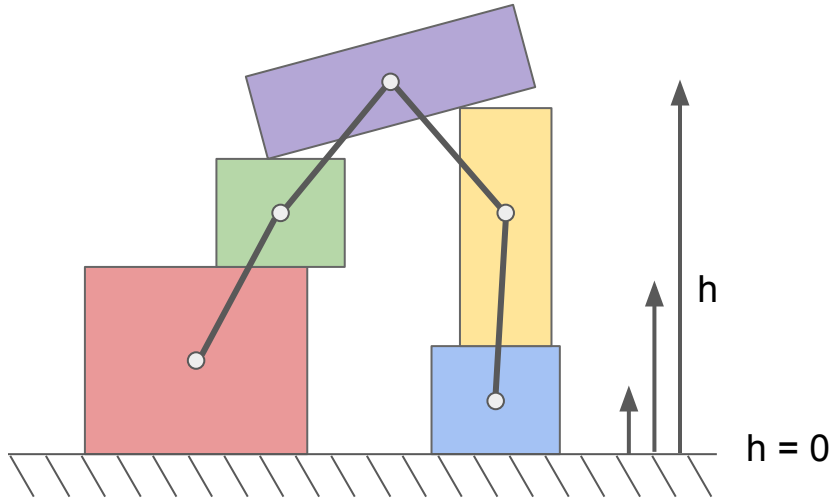


Stiff stack



$$s_i(x_i) = e^{-kh(x_i)} \quad k \in [1, 5]$$

$$m_i^* = s_i m_i \quad \text{scaled mass for 1 iteration}$$



Better convergence rate

Less pressure on lower object in stack

Create matter out of nowhere

Conclusion

Everything is particles

```
struct Particle {  
    float x[3];  
    float v[3];  
    float inv_mass;  
    int phase;  
};
```

Collisions

Solid $\Rightarrow$ Fluid

Fluid $\Rightarrow$ Solid

SDF Collisions

Position based Friction

Some tricks

Stiff stack

Biased Jacobi

Const size particles

Particle Simulation using CUDA (2010) [Loïs GDL]

Uniform grids =

